

weighing just 1.2kg, presented only a 10 per cent risk of severe injury. The largest drone, weighing more than 11kg, presented a 70 per cent risk.

The analysis—published in the journal *Annals of Biomedical Engineering*—suggests injury risks can be mitigated if drones are designed to fracture and break upon impact.

“If you reduce the energy that’s able to be transferred to be head, you reduce the injury risk,” said doctoral student Eamon Campolettano. “The overarching goal for manufacturers should be to limit energy transfer.”

While some smaller drones are already safe enough to take on commercial duties, the study was more about establishing a baseline for how to carry out risk assessments for drone-human impacts.

Zero-power skin patches capable of data transfer up to 2.8km

University of Washington (UW) researchers have demonstrated almost zero-power devices that can transmit data across distances of up to 2.8 kilometres—enabling a vast array of interconnected devices.



The UW research team built this flexible epidermal patch prototype that successfully transmitted information across a 306sq.m (3300sq.ft) atrium (Image courtesy: Dennis Wise/University of Washington)

Until now, devices that can communicate over long distances have consumed a lot of power. The UW team’s latest long-range backscatter system provides reliable long-range communication with sensors that consume 1000 times less power than existing technologies capable of transmitting data over similar distances. It’s an important and necessary breakthrough toward embedding connectivity into billions of everyday objects.

The sensors are so cheap—with an expected bulk cost of 10 to 20 cents each—that farmers looking to measure soil temperature or moisture could affordably blanket an entire field to determine how to efficiently plant seeds or water. Other potential applications range from sensor arrays that could monitor pollution, noise or traffic in ‘smart’ cities or

medical devices that could wirelessly transmit information about a heart patient’s condition around the clock.

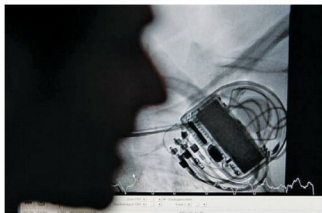
The system has three components: a source that emits a radio signal, sensors that encode information in reflections of that signal and an inexpensive off-the-shelf receiver that decodes the information. When the sensor is placed between the source and receiver, the system can transmit data at distances up to 475 metres. When the sensor is placed next to the signal source, the receiver can decode information from as far as 2.8 kilometres away.

Three reasons why pacemakers are vulnerable to hacking

The US Food and Drug Administration (FDA) recently recalled approximately 465,000 pacemakers made by the company Abbott’s. The reason for the recall? The devices can be remotely ‘hacked’ to increase activity or reduce battery life, potentially endangering patients.

James H. Hamlyn-Harris, senior lecturer, computer science and software engineering, Swinburne University of Technology, explains, “The qualities that make remotely-accessible human implants desirable—namely, low cost, low-maintenance batteries, small size, remote access—also make securing such devices a serious challenge.”

“Three key issues hold back cyber-safety: Most embedded devices don’t have the memory or power to support proper cryptographic security, encryption or access



Embedded medical devices will continue to be vulnerable to cybersecurity threats. The pacemaker depicted is not made by Abbott’s (Image courtesy: REUTERS/Fabrizio Bensch)

control. Doctors and patients prefer convenience and ease of access over security control. Remote monitoring, an invaluable feature of embedded devices, also makes them vulnerable,” he adds.

Conventional cryptography suites (algorithms and keys used to prove identity and keep transmissions secret) are designed for computers, and involve complex mathematical operations beyond the power of small, cheap Internet